## 1. The round and the comparator

### 1. Full-information prediction

There are $n \geq 1$ experts and $T \geq 1$ rounds. Expert $i$ is an advice rule. Each round follows this order.

1. Read the experts' predictions and commit to a decision.
2. Observe the outcome. The learner incurs one scalar loss $L_t(\mathcal{A})$. The full vector $\ell_t$ gives every expert's loss $\ell_t(i)$.
3. Use the revealed losses to prepare the next decision.

With binary labels, zero-one loss is $1$ for a mistake and $0$ for a correct prediction.

### 2. External regret

First sum each expert's losses over the whole sequence. Let $L^*$ be the smallest total.

$$L_i = \sum_{t=1}^{T} \ell_t(i), \quad L^* = \min_i L_i.$$

The **best fixed expert** uses the same expert ID on every round. External regret measures the gap.

$$R_T = \sum_{t=1}^{T} L_t(\mathcal{A}) - L^*.$$

Taking a new minimum on each round would allow the comparator to switch experts.

### 3. Average excess loss

$\frac{R_T}{T}$ measures excess loss per round. **Sublinear regret** means $\frac{R_T}{T} \to 0$. A sublinear upper bound $R_T \leq b(T)$ with $\frac{b(T)}{T} \to 0$ lets the ratio stay negative when the learner consistently beats every fixed expert.

If all experts lose $1$ on every round, following any of them gives zero regret and average loss $1$.

## 2. Halving with a perfect expert

### 1. Vote, then keep the correct experts

A **perfect expert** predicts every outcome correctly over the stated horizon. Start with all $n$ experts in the active set $S_1$.

1. Predict the majority binary label among experts in $S_t$. Choose $0$ in a tie.
2. After seeing the true label, keep exactly the active experts that predicted it. They form $S_{t+1}$.

Update even when the learner predicted correctly. A mistaken minority expert still leaves the set.

### 2. Count survivors

Let $M$ be the learner's number of mistakes. On a mistake, at least half the active experts voted for the wrong label and leave. On any other round, the set can only stay the same or shrink.

A perfect expert survives all $T$ rounds. Together these facts give

$$1 \leq |S_{T+1}| \leq \frac{n}{2^M},$$

$$M \leq \lfloor \log_2 n \rfloor.$$

The bound counts learner mistakes across all $T$ rounds.

### 3. When the perfect-expert premise fails

The expert with the fewest mistakes can still have made one. If every expert has been wrong, the active set is empty and there is no majority to follow. Positive weights let an expert contribute after a mistake.

## 3. Deterministic weighted majority

### 1. Vote by weight

Use binary labels, set $w_1(i) = 1$ for each expert and fix $0 < \eta \leq \frac{1}{2}$. Add up the weights voting for each label. Predict the label with greater weight, choosing $0$ in a tie.

After the outcome, multiply each mistaken expert's weight by $1 - \eta$. Correct experts keep their weights. Apply this update on every round.

### 2. Compare mistake counts

Let $M_i$ count expert $i$'s mistakes and let $M^* = \min_i M_i$. The total weight is $\Phi_t = \sum_i w_t(i)$. On a learner mistake, at least half the weight is wrong, so $\Phi_{t+1} \leq (1 - \frac{\eta}{2})\Phi_t$. Expert $i$ retains weight $(1-\eta)^{M_i}$. Thus

$$(1-\eta)^{M_i} \leq \Phi_{T+1} \leq n\Big(1 - \frac{\eta}{2}\Big)^M.$$

For the stated rate this gives

$$M \leq 2(1+\eta)M^* + \frac{2\log(n)}{\eta}.$$

Here and below, $\log$ is the natural logarithm.

### 3. Why deterministic regret can be linear

Take two constant experts, one predicting $0$ and the other $1$. An opponent can infer a deterministic learner's next label and choose the opposite outcome. The learner then makes $T$ mistakes. Exactly one expert is wrong each round, so

$$M^* \leq \frac{T}{2}, \quad R_T = M - M^* \geq \frac{T}{2}.$$

## 4. Private sampling and mixture loss

### 1. Randomized weighted majority

Keep the binary weight update, equal starting weights and fixed $0 < \eta \le \frac{1}{2}$. Before the outcome, normalize the weights and draw one expert $I_t$. Follow that expert's advice.

$$p_t(i) = \frac{w_t(i)}{\Phi_t}, \quad I_t \sim p_t.$$

After the outcome, update all experts, including those the learner did not draw.

### 2. What the opponent may see

A **nonanticipating adversary** may use past draws and the public probabilities $p_t$. It fixes the current loss vector without seeing the current draw and reveals it after commitment. An adversary that fixes the entire sequence in advance also satisfies this condition.

### 3. One draw versus its mean

A sampled expert incurs the coordinate loss $\ell_t(I_t)$. The **mixture loss** is the weighted average

$$\langle p_t, \ell_t \rangle = \sum_i p_t(i)\ell_t(i).$$

Given the history, $p_t$ and the fixed $\ell_t$, this average equals the conditional expected loss of the draw. For binary losses it is the probability of drawing a mistaken expert. For four equally weighted experts with losses $(1, 1, 0, 0)$, the mixture loses $\frac{1}{2}$. A sampled expert loses either $0$ or $1$.

## 5. Linear multiplicative weights

### 1. Bounded losses and the update

Use $n \ge 1$ experts, $w_1(i) = 1$, and fixed $0 < \eta \le \frac{1}{2}$. Normalize to $p_t$ and commit. The full vector $\ell_t \in [0, 1]^n$ then determines the next weights.

$$w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i)).$$

Each factor lies in $[1 - \eta, 1]$, so weights stay positive. At losses $0$ and $1$, this gives the binary weight update.

### 2. Two bounds on the same total weight

For a mixture decision, write $L_t = \langle p_t, \ell_t \rangle$. Summing the updated weights gives the exact identity

$$\Phi_{t+1} = \Phi_t(1 - \eta L_t).$$

Using $1 - z \le \exp(-z)$ over all rounds gives

$$\Phi_{T+1} \le n \exp\left(-\eta \sum_{t=1}^{T} L_t\right).$$

Each expert supplies a lower bound. For $0 \le x \le 1$, $(1 - \eta)^x \le 1 - \eta x$. Multiplying these bounds gives

$$\Phi_{T+1} \ge w_{T+1}(i) \ge (1 - \eta)^{L_i}.$$

### 3. From total loss to regret

Using $\log(1 - \eta) \ge -\eta(1 + \eta)$ and combining the bounds yields, for each fixed expert $i$,

$$\sum_{t=1}^{T} L_t \le (1 + \eta)L_i + \frac{\log(n)}{\eta}.$$

Choose the best fixed expert and subtract $L^*$. The mixture regret on every sequence satisfies

$$R_T \le \frac{\log(n)}{\eta} + \eta L^* \le \frac{\log(n)}{\eta} + \eta T.$$

## 6. Applying the regret guarantee

### 1. Expected regret for a sampled expert

Under nonanticipation, summing the conditional means gives the same expected cumulative loss for the draws and the mixture. If later losses use earlier draws, $L^*$ can itself be random. Compare on the same sequence.

$$E[R_T] = E\left[\sum_{t=1}^{T} \ell_t(I_t) - L^*\right]$$

$$= E\left[\sum_{t=1}^{T} \langle p_t, \ell_t \rangle - L^*\right] \le \frac{\log(n)}{\eta} + \eta T.$$

Keep $L^*$ inside the expectation. Individual sampled runs can have different regret.

### 2. Choose the rate for a known horizon

For $n \ge 2$, choose a single rate for the whole run.

$$\eta = \min\left\{\frac{1}{2}, \sqrt{\frac{\log(n)}{T}}\right\}.$$

**When** $T \ge 4\log(n)$, the two terms in the coarse bound are equal. The resulting bound is $2\sqrt{T\log(n)}$ for mixture regret or expected sampled regret.

**When** $T < 4\log(n)$, use $\eta = \frac{1}{2}$. The coarse bound is $2\log(n) + \frac{T}{2}$. The trivial cap $R_T \le T$ also applies.

### 3. One expert or an unknown horizon

With $n = 1$, the learner always uses the single expert and $R_T = 0$. Any legal positive rate works.

For fixed $n$ and large known $T$, the bound per round is $2\sqrt{\frac{\log(n)}{T}}$. Each run uses its own fixed rate. A rate that changes during a run needs a separate analysis.